

Ernesto Luis Malta Rodrigues Filho

Software Engineer | C++, Java, Python | Azure, OpenShift, Kubernetes | High Performance, Embedded Systems
ernesto.filho.3942@gmail.com | +55 41 99630-1739 | linkedin.com/in/elmrf | github.com/Lohkdesgds | Paraná, Brazil (GMT-3) |
Available for Remote Work | EU & US

Summary

Software Engineer with 4 years of experience in the energy sector building scalable Java micro-services on Azure and OpenShift, and extensive (9+ years) academic and project-based experience in C/C++ for embedded systems and high-performance applications. Accustomed to working with multinational teams, across the USA, Thailand, India, and others. Delivered a configurable data access service that removed the need for client teams to build and maintain their own microservices. Developed and maintains C++ libraries and projects for desktop and embedded platforms, looking for new challenges in high performance, low level, reusable, and maintainable development in this domain.

Experience

Software Engineer

ExxonMobil | Paraná, Brazil

01/2023 – 09/2026

Developed a centralized, configurable data-access service on a multinational team on Azure Red Hat OpenShift (ARO) that enables clients to deploy customized API endpoints through low-code YAML configurations in GitHub, providing self-service access to Snowflake data without requiring them to build and maintain their own microservices.

- Reduced costs by around **\$30k+** per client by centralizing resources on this one-size-fits-all solution, which freed multiple namespace instances costs on Azure, and enabled a more standardized, secure, and performant solution.
- Engineered the supporting CI/CD automation using GitHub Actions, Python, and Bash to validate configurations and streamline deployment of customer-specific service instances.
- Designed the one-size-fits-all API automation approach after researching multiple languages and paradigms, and mentored the distributed USA, Thailand, India and Brazil team through delivery.
- Led improvements to production observability by designing custom telemetry, Datadog monitors, and Snowflake dashboards, providing real-time visibility into pipeline health and operational failures.
- Spearheaded cross-platform build standardization across Windows, MacOS and Linux, eliminating the need for additional hardware and avoiding **\$120k+ in costs**.

Skills: Java, Spring Boot, Cloud, Azure, TDD, Redis, REST, SQL, Snowflake, JWT, OAuth2, Kubernetes, k6, CI/CD, Git, Python, GitHub Actions, Datadog, Windows, Linux

Teaching Assistant – Fundamentals of Programming

UTFPR | Paraná, Brazil

02/2018 – 12/2019

Provided technical guidance and troubleshooting support for undergraduate C/C++ programming coursework. Guided students to follow better standards, helped them installing required software for the class, and helped teacher with students' issues.

- Hosted and maintained an educational portal to assist students with compiler setup, library configuration, and code debugging. With extensive guidelines, documentation, and step-by-step tutorials, the portal enabled students to start developing on day one.
- Guided students set up and troubleshoot issues in their code, mentoring them to follow design patterns and standards, enabling them to conclude their homework faster, and with more confidence.

Skills: C, C++, CMake, PHP, HTML, CSS, JavaScript, SQL, Git

Projects & Open Source

Modular Monitor Project

Capstone Project – github.com/ModularMonitor

01/2024 – 10/2024

Developed a modular real-time monitoring engine for ESP32 in C++ with hot-swappable I²C sensor architectures, multithreading, and custom device drivers.

- Developed drivers for hot-swappable Serial through I²C communication, custom TCP socket design for wireless monitoring with REST API, and robust SD card streaming of CSV data.
- Enhanced open source TFT display library for responsive touchscreen user interface, 4G GSM connection, Circular Queue, and multiple Adafruit sensor libraries, to fit constrained hardware.
- Enabled multi-sensoring multi-module design on Espressif ESP32 platform by leveraging multiple hardware-specific tweaks.

Skills: C++, ESP-IDF, Arduino, CMake, Shell, Bash, CI/CD, Git, REST

Lunaris Library Collection

github.com/LunarisLib

10/2020 – Present

Authored a cross-platform, open source, utility library featuring lock-free thread primitives, asynchronous I/O, socket abstraction, encryption, and a memory-optimized JSON parser based on cJSON.

- Engineered a suite of 14+ modern C++ modules to accelerate application development, centralizing core features to eliminate code duplication across multiple projects
- Implemented robust build and testing pipelines using CMake and CTest, which reduced compilation times, standardized workflows, and simplified library linking for dependent applications.
- Created automated version release pipelines to enable the continuous deployment of libraries, simplifying dependency linking for downstream applications.

Skills: C++, CMake, Git, GitHub Actions, Shell, Bash, CI/CD, Windows, Linux, Automated Testing, Clean Architecture

Personal website

github.com/Lohkat – lohkat.github.io

06/2023 – Present

Developed and deployed a responsive web portfolio made with modern HTML5, CSS3, and JavaScript, utilizing GitHub Pages as backend with automated deployment and modular design.

- Built interactive frontend features – dynamic galleries, web-player, and search tools – showcasing cross-domain technical adaptability.
- Architected modular design by utilizing multiple repositories, spreading weight and deployments for different parts of the website, accomplishing performance on updates and releases.

Skills: HTML, CSS, JavaScript, Git, GitHub Actions, CI/CD

Education

M.S. in Electrical Engineering & Industrial Informatics (CPGEI)

01/2026 – Present

Universidade Tecnológica Federal do Paraná – Paraná, Brazil

B.S. in Electrical and Electronics Engineering

01/2016 – 10/2024

Universidade Tecnológica Federal do Paraná – Paraná, Brazil

Languages

English: Full professional proficiency.

Portuguese: Native

Skills

Languages: C++, C, Java, Shell / Bash, Python, JavaScript / TypeScript, HTML, CSS

Systems & Infrastructure: Linux, Windows, FreeRTOS, Azure, Red Hat OpenShift, Docker, Kubernetes, GitHub Actions

Tools, Frameworks & Practices: CMake, Git, Spring Boot, ESP-IDF, Arduino, REST, Snowflake, Redis, Datadog, k6, CI/CD, TDD (Test Driven Development), Event Driven, Agile / Scrum, Clean Code , automated testing